DAY 1 OF 30

What is Microsoft Fabric, really?

The big picture first.
Then what matters in production.

OneLake | Delta tables | Capacity

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THE PROBLEM

One data platform used to mean six separate services.

1	**Move the data**	Azure Data Factory
2	**Store it**	Data Lake Storage
3	**Clean it**	Spark on Databricks or Synapse
4	**Query it**	A SQL warehouse
5	**Live data**	Event Hubs plus a streaming engine
6	**Show it**	Power BI

Six setups. Six bills. Six security models. **And you do the wiring between them.**

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THE SIMPLE VERSION

Fabric is a fitted kitchen.

Before, you bought the oven, fridge and sink from different shops and did the plumbing yourself.

Fabric arrives as **one kitchen**. Every appliance is already connected, and they all use **one pantry**.

✓ **One sign-in**	It runs in the browser, like Power BI
✓ **One storage**	Called OneLake. Data is kept once
✓ **One bill**	A single pool of computing power

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THE MAP

What is inside Fabric

Data Factory
Bring data in, on a schedule

Data Engineering
Clean it with Spark notebooks

Data Warehouse
Query it with plain SQL

Real-Time Intelligence
Data that arrives every second

Data Science
Train and track ML models

Power BI
Reports people actually open

OneLake
One copy of the data, saved as Delta tables. Every tool above reads the same files.

Capacity · one shared pool of compute that all of it runs on

Security and governance · set once, applies everywhere

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FOR ENGINEERS

What changes for you

It is SaaS, not PaaS

No clusters to build or patch. Spark starter pools give you a session in seconds.

One open format: Delta Parquet

Spark writes a table. The SQL endpoint and Power BI read that same table. No export step.

One compute currency

Every workload spends Capacity Units from the same F SKU. Sizing is one decision, not six.

Shortcuts instead of copies

Point OneLake at ADLS, S3 or Google Cloud Storage and query the data where it sits.

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THE HONEST PART

Is Fabric right for you?

GOOD FIT

Your company already lives in Power BI and Azure

A small team that wants less infrastructure work

You need batch, real-time and reports in one place

THINK TWICE

You need fine control over cluster configuration

You are deep in another platform that works well

Nobody will watch capacity usage

> **The catch people miss:** capacity is shared. One heavy Spark job can slow down everybody's reports. Plan your capacity before you plan your pipelines.

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FROM REAL WORK

What this looked like on a real project

A people-analytics platform. Employee data lived in scattered CSV files that were updated by hand.

1. **One Lakehouse on OneLake** replaced the scattered files
2. **Triggered pipelines** refresh the data daily, no manual step
3. **Bronze, Silver, Gold** layers built in PySpark
4. **A semantic model** on top for the organisation's KPIs
5. **Row-level security** so people see only what they should

All five steps, **inside one product**.

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SAVE THIS

Fabric in five lines

1. **One platform** for the whole data journey
2. **OneLake** stores the data once, as Delta tables
3. **Every engine** reads those same tables
4. **One capacity** pays for everything, so watch it
5. **Less plumbing**, more time on the real problem

Tomorrow: OneLake.
Why "store it once" is a bigger idea than it sounds.

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